

Save As

Copy Figure

Print

Show Code

Title

Subtitle

X-Label

Y-Label

Z-Label

Legend

Colorbar

Grid

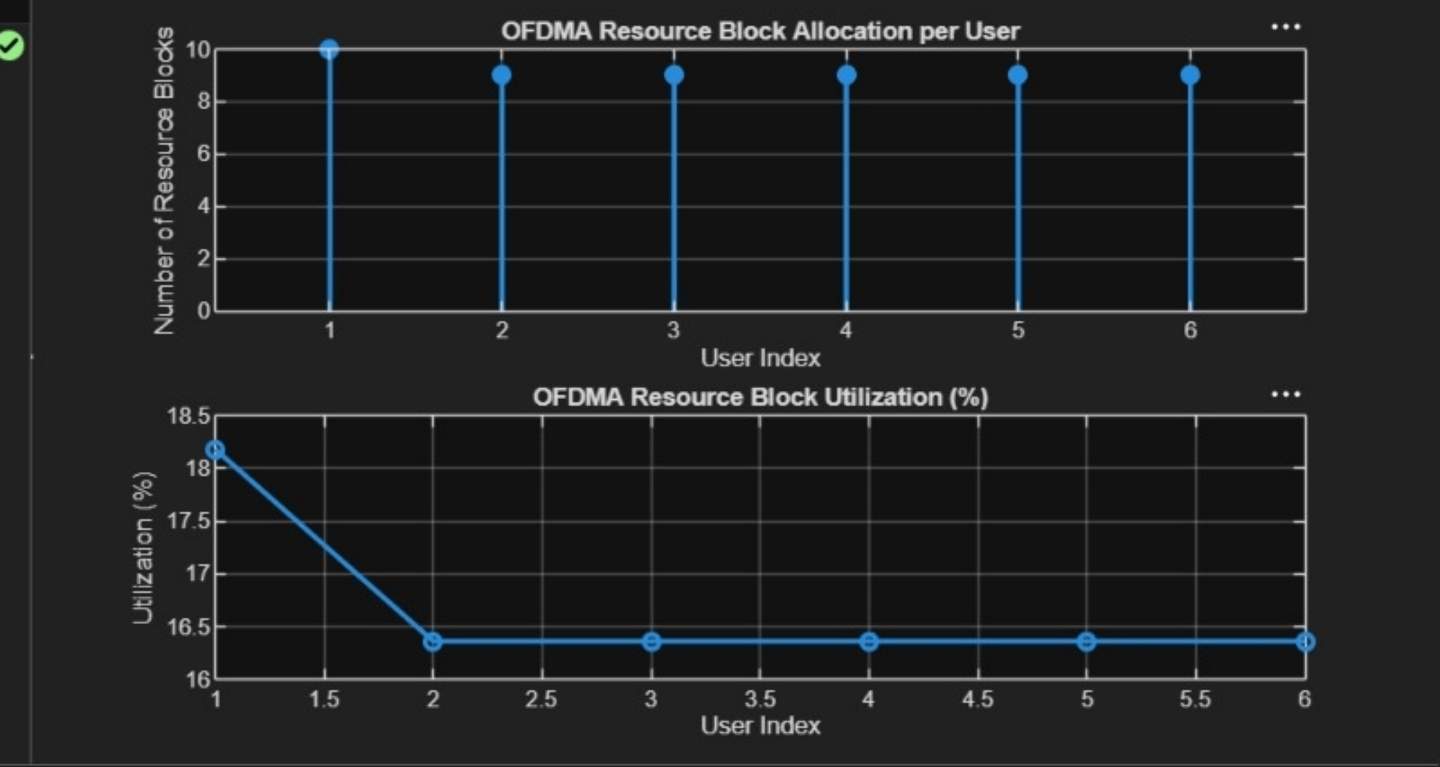
X-Grid

Y-Grid

FILE LABELS AND ANNOTATIONS

```

9 % RB allocation (simple equal + remainder distribution)
10 base = floor(total_RB / length(users));
11 alloc = base * ones(1,length(users));
12 alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users))) + 1;
13 utilization = (alloc / total_RB) * 100;
14 disp('Total Resource Blocks:')
15 disp(total_RB)
16 disp('RB Allocation per User:')
17 disp(alloc)
18 figure
19 % ----- Graph 1: RB Allocation per User -----
20 subplot(2,1,1)
21 stem(users, alloc, 'filled', 'LineWidth', 2)
22 title('OFDMA Resource Block Allocation per User')
23 xlabel('User Index')
24 ylabel('Number of Resource Blocks')
25 grid on
26 % ----- Graph 2: RB Utilization -----
27 subplot(2,1,2)
28 plot(users, utilization, '-o', 'LineWidth', 2)
29 title('OFDMA Resource Block Utilization (%)')
30 xlabel('User Index')
31 ylabel('Utilization (%)')
    
```



Total Resource Blocks:
 55

RB Allocation per User:
 10 9 9 9 9 9

>>

Copy Figure

Print

Show Code

Title

Subtitle

X-Label

Y-Label

Z-Label

Legend

Colorbar

Grid

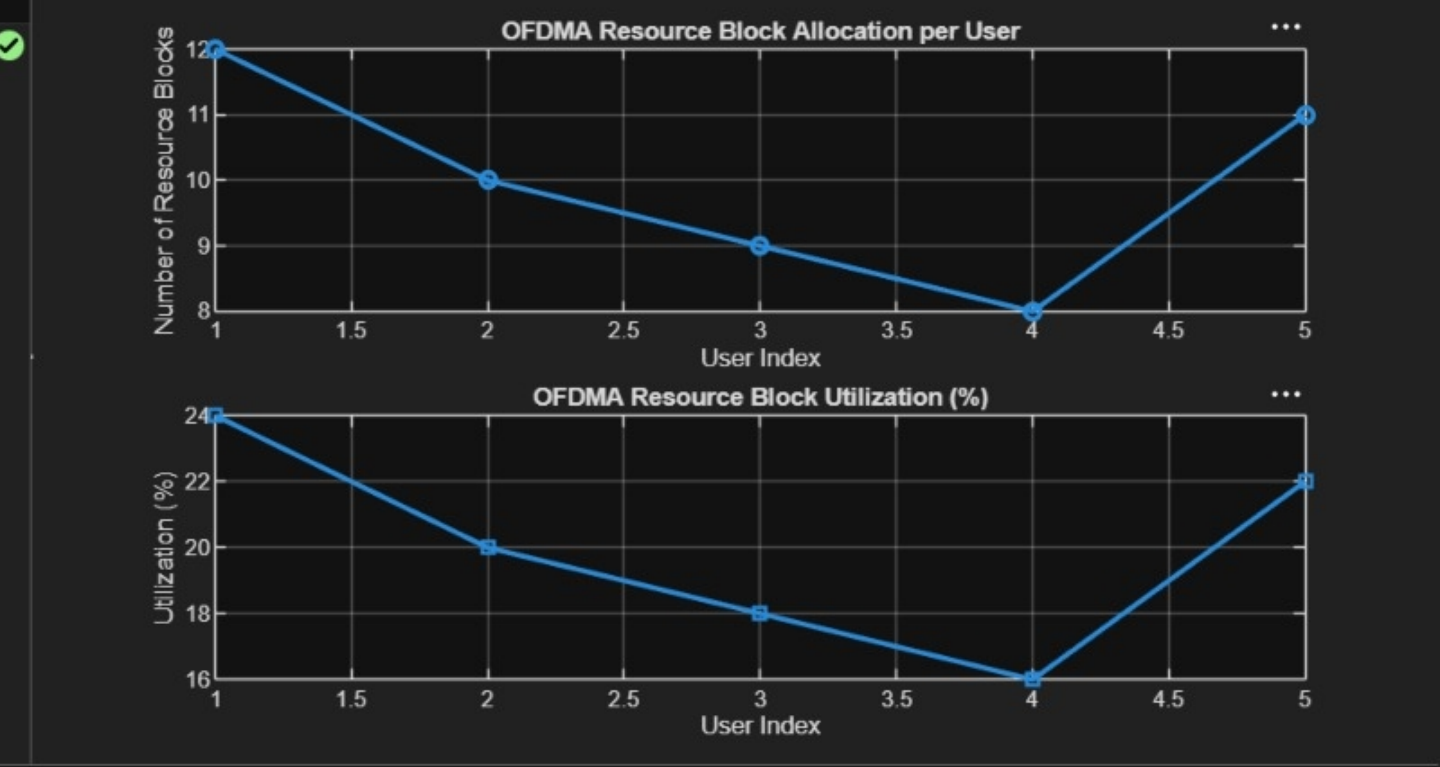
X-Grid

Y-Grid

LABELS AND ANNOTATIONS

```

3 total_RB = 50;
4 % Number of active users
5 users = 1:5;
6 % RB allocation to users (example scenario)
7 alloc = [12 10 9 8 11];
8 % Resource Block Utilization (%)
9 util = (alloc / total_RB) * 100;
10 disp('Resource Block Utilization (%) per User:')
11 disp(util)
12 figure
13 % ----- Graph 1: RB Allocation -----
14 subplot(2,1,1)
15 plot(users, alloc, '-o', 'LineWidth', 2)
16 title('OFDMA Resource Block Allocation per User')
17 xlabel('User Index')
18 ylabel('Number of Resource Blocks')
19 grid on
20 % ----- Graph 2: Utilization (%) -----
21 subplot(2,1,2)
22 plot(users, util, '-s', 'LineWidth', 2)
23 title('OFDMA Resource Block Utilization (%)')
24 xlabel('User Index')
25 ylabel('Utilization (%)')
    
```



Command Window

```

24
20
18
16
22
>>
    
```

Save As

Copy Figure

Print

Show Code

Title

Subtitle

X-Label

Y-Label

Z-Label

Legend

Colorbar

Grid

X-Grid

Y-Grid

FILE LABELS AND ANNOTATIONS

```

4 users = 1:5;
5 % Allocated Resource Blocks per user
6 alloc_RB = [12 10 9 8 11];
7 % Modulation efficiency (bits/symbol)
8 mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
9 % Throughput calculation (Mbps)
10 throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
11 disp('User Throughput (Mbps):')
12 disp(throughput')
13 figure
14 % ----- Graph 1: Throughput per User -----
15 subplot(2,1,1)
16 plot(users, throughput, '-o', 'LineWidth', 2)
17 title('OFDMA User Throughput')
18 xlabel('User Index')
19 ylabel('Throughput (Mbps)')
20 grid on
21 % ----- Graph 2: RB vs Throughput -----
22 subplot(2,1,2)
23 plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
24 title('Resource Blocks vs Throughput Relationship')
25 xlabel('Allocated Resource Blocks')
26 ylabel('Throughput (Mbps)')
    
```

Command Window

```

4.3200
7.2000
9.7200
5.7600
3.9600
>>
    
```

